Scalability and Future Plan

AI/ML and Generative AI Track • Project Documentation

Project	Trip Genius AI (VoyagePlan Travel Planner)	**Date**	26 October 2026
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Current baseline

The application is deployed with a Vercel frontend, a Render FastAPI backend, Groq for generation and Supabase for authentication and data. The production smoke tests pass for the core workflow. Throughput, latency and concurrency limits have not been measured, so current capacity is not asserted.

Scalability risks and mitigations

Area	Potential risk	Recommended action
AI provider (Groq)	Rate limits, variable latency, token cost	Set timeouts, retry only safe transient errors, rate-limit users, monitor usage.
Backend hosting (Render)	Cold starts or resource limits	Measure latency; review Render metrics and service plan.
Database (Supabase)	Unindexed queries or policy mistakes	Review query patterns, indexes and RLS policies.
API	Burst traffic and duplicate requests	Add request and input limits; cache only where safe.
Reliability	Provider or database outage	Return actionable errors, log request IDs, define recovery steps.
Security	Secret exposure or unauthorised trip access	Keep secrets server-side, rotate exposed keys, test RLS with two accounts.

Roadmap

Phase	Planned improvement	Acceptance evidence
1	Security and configuration audit (RLS, cross-account test)	No committed secrets; access policies reviewed
2	Automated tests for API and saved-trip workflow	Repeatable test report in repository / CI
3	Performance baseline and controlled load test	Actual latency, error-rate and resource metrics
4	Observability and provider-failure handling	Logs, timeouts, useful user messages
5	Structured itinerary schema, PDF export, shareable links	Consistent rendering; export works
6	Optional maps, budget breakdown and verified travel data	Documented API source, freshness and limitations
7	Accessibility and UX review	Keyboard, contrast and responsive checks

Success criteria

Set measurable targets only after baseline tests. Future enhancements must not compromise the working flow, user-data privacy, or the reliability of saving and editing trips.